

Debashis Pani

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Education

B.Tech in Information Technology, Kalinga Institute Of Industrial Technology
CGPA: 8.49 , Graduation Year: 2026

Experience

SDE Intern, Dfy-Graviti

Jan 2026 – Present

- Developing scalable backend systems for geospatial and real-time tracking applications using **FastAPI**, **Django**, **Laravel**, **PostgreSQL**, **TimescaleDB**, **Docker**, and Linux infrastructure.
- Managing and optimizing TimescaleDB pipelines handling 50M+ AIS records for real-time maritime tracking and analytics.
- Orchestrated the migration of over 15 backend APIs from Django and Laravel to FastAPI, enhancing API response times by ~40% and reducing server resource usage.
- Built and optimized REST APIs and backend workflows processing over 1M+ database records, improving scalability, reducing query latency, and ensuring reliable high-volume data handling.

SDE Intern, NTRO(National Technical Research Organisation)

May 2025 – Aug 2025

- Developed scalable production-grade modules using **React**, **FastAPI**, **Laravel**, **PostgreSQL**, and **Docker**.
 - Automated CI/CD deployment pipelines using **Docker** and **Python** on **Linux-based servers**, reducing manual deployment effort by ~50%.
 - Designed and maintained containerized environments to improve service isolation, reliability, and long-term maintainability.
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Achievements

Smart India Hackathon 2025 – Winner (Tech Lead)

- Led a team of **6** to build a **Multimodal RAG system** with **offline LLM inference** using **Ollama**, supporting PDF, DOCX, PPT, and audio inputs, competing against **20 teams** in the national finale.
 - Implemented **semantic search** using embeddings and real-time streaming responses via **FastAPI + Next.js**.
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Projects

Patient Management System (Microservices)

Developed a scalable healthcare microservices platform using **Spring Boot**, **Kafka**, **gRPC**, **Docker**, and **PostgreSQL** to manage patient records, billing, and analytics workflows.

Code: github.com/pani2004/Patient-Management-System

CodeLens AI – Codebase Analysis and Understanding Tool

Built an AI-powered platform to analyze and understand large codebases using **FastAPI**, **LangChain**, **Next.js**.

- Enabled natural language interaction with repositories using **Gemini AI**.
- Implemented **semantic search** using **pgvector embeddings** for efficient code retrieval.

Code: github.com/pani2004/CodeLens

OneBox Email Aggregator (AI + Real-time Sync)

AI-powered email aggregation platform built with **TypeScript**, **Node.js**, **React**, **PostgreSQL**, and **Docker**.

- Engineered a hybrid search pipeline by combining **Elasticsearch** full-text indexing with **ChromaDB** semantic retrieval, improving search relevance and accelerating data discovery by **30%**.
- Implemented AI workflows using **Gemini API** for email classification and RAG-based reply suggestions.

Code: github.com/pani2004/OneBox

Portfolio Using React & Tailwind CSS

Personal portfolio website showcasing projects and skills.

Demo: portfolio-beta-liart-56.vercel.app Code: github.com/pani2004/Portfolio

Technical Skills

Languages: C++, Java, JavaScript, TypeScript, Python

Frameworks: Next.js, React, Tailwind CSS, Node.js, FastAPI, Spring Boot, Django, Laravel, LangChain, Vue.js

Databases: PostgreSQL, MySQL, MongoDB, Prisma

Tools: Linux, Docker, Git, GitHub, Kafka, Redis, AWS EC2, gRPC, Elasticsearch, CI/CD